

⚠ This quiz has been regraded; your new score reflects 2 questions that were affected.

Quiz 3

Mar 26, 2021

Instructions

If you have questions - ping me on discord or connect to my office hours zoom:

<https://ksu.zoom.us/j/95088882288> <https://ksu.zoom.us/j/95088882288>

Attempt History

Attempt

Time

Score

Regraded

Question 1

3 / 3 pts

If you were to replace Tesla with other units, the following combination would work:

☐ C · m/s

Correct!

☐ $C \cdot s/m$ ☐ C/kg ☒ $kg/(C \cdot s)$ ☐ $N/(C \cdot m)$ **Question 2****3 / 3 pts**

If you have a charged particle in a magnetic field, then the field exerts a force on it

☐ always☐ never☒ if the particle is moving across the field lines☐ if the particle is moving along the field lines☐ if the particle is at rest

Correct!

Question 3**3 / 3 pts**

You have a simple electron gun-target experiment: a beam of electrons that shoots out from a small window in a horizontal plane aimed directly at a target some distance away. If the electrons enter an area with a magnetic field directed upward on the way to the target, the electrons will deflect

Correct!

- ☐ upward
- ☐ downward
- ☒ to the left of the target
- ☐ to the right of the target
- ☐ they will not get deflected at all

Question 4 Original Score: 3 / 3 pts **Regraded Score: 3 / 3 pts**

⚠ This question has been regraded.

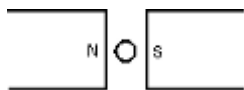
Find the true statement about a static magnetic field:

Correct!

- ☐ it cannot exert a force on a charge
- ☐ it cannot accelerate a charge
- ☐ it cannot change the momentum of the charge
- ☒ it cannot change the kinetic energy of a charge
- ☐ it cannot exist

Question 5**3 / 3 pts**

If you have a particle with an electric charge of $4.8 \times 10^{-19} \text{ C}$ that enters in the uniform electric field of $5 \times 10^4 \text{ V/m}$ perpendicular to a uniform magnetic field of 0.5 T and you find that the particle's acceleration is exactly zero, then its speed must be

☐ 0 m/s☐ $1.6 \times 10^{-19} \text{ C}$ ☐ $2.5 \times 10^5 \text{ m/s}$ ☐☒ 10^5 m/s ☐ Any value but 0 m/s**Correct!****Question 6****3 / 3 pts**

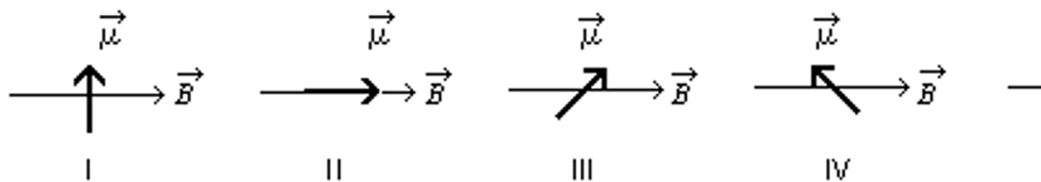
In the diagram above, the round circle is a straight wire that carries a current toward the viewer (aka out of page or out of screen). The wire is positioned between the poles of a permanent magnet indicated by letters N and S. The direction of the magnetic force that the magnet exerts on the wire is:

☒ up☐ down☐ to the left**Correct!**

- ☐ to the right
- ☐ into the page/screen
- ☐ out of the page/screen

Question 7

3 / 3 pts



Which of the following orientations of a magnetic dipole $\vec{\mu}$ in a uniform magnetic field $\vec{B}$ would have the lowest potential energy?



☐ I

☒ II

☐ III

☐ IV

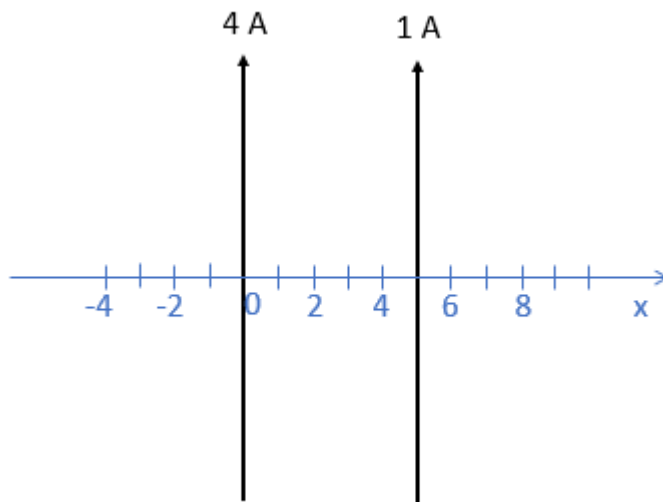
☐ V

Correct!

Question 8

3 / 3 pts

Consider a long straight wire that produces a magnetic field of $2 \times 10^{-5} \text{ T}$ at a distance of 2 cm from the wire. The current in the wire is

Correct!☐ 0.16 A☐ 1.0 A☒ 2.0 A☐ 4.0 A☐ 25 A**Question 9****3 / 3 pts**

The drawing above depicts two currents, 4 A and 1 A, that flow in the same direction along very long straight wires. The x-coordinates of these currents are 0 and 5, respectively (no need for units here). Find an x-coordinate where the magnetic field from the wires vanishes:

☐ there is no such point with the given configuration☐ 6☒ 4**Correct!**

☐ 2☐ -2☐ None of the above**Question 10** Original Score: 3 / 3 pts **Regraded Score: 3 / 3 pts****! This question has been regraded.**

A proton and an α -particle ($m_\alpha/m_p = 4$, $Q_\alpha/Q_p = 2$) have the same momentum and move in a constant B -field at right angles to B -field direction. How will the radius of the α -particle circle compare to that of the proton?

☐ $R_\alpha/R_p = 1$ ☐ $R_\alpha/R_p = 2$ ☐ $R_\alpha/R_p = 4$ ☒ $R_\alpha/R_p = 0.5$

Correct Answer

You Answered

☐ $R_\alpha/R_p = 0.25$ **Question 11****3 / 3 pts**

Consider two infinite straight wires each carrying equal currents of 1 A in the opposite directions. They repel each other with a force of 1 N. If you

double the currents, the repulsion force will be

☐ 0.25 N

☐ 0.5 N

☐ 1 N

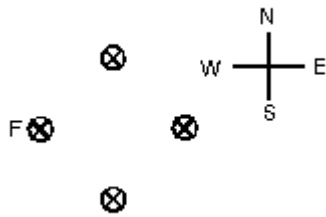
☐ 2 N

☒ 4 N

Correct!

Question 12

3 / 3 pts



The image above shows a cross-sectional view of four wires, each carrying an identical current into the screen. The magnetic force on the wire marked F is directed

☐ up

☐ down

☐ north

☐ south

☒ east

Correct!

☐ west

Question 13

3 / 3 pts

In the $\oint \vec{B} \cdot d\vec{s} = \mu_0 i$ the integration must be over

☐ surface

☐ closed surface

☐ path

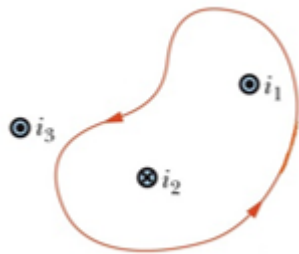
☒ closed path

☐ closed path that surrounds all the current producing magnetic field $\vec{B}$

Correct!

Question 14

3 / 3 pts



In the figure above, there are three wires with $i_1 = 1$ A, $i_2 = 10$ A, and $i_3 = 4$ A carrying currents out of, into, and out of the screen ("page") respectively. You can also see an Amperian loop in red with a direction of integration indicated by arrows. The value of the $\frac{1}{\mu_0} \oint \vec{B} \cdot d\vec{s}$ is

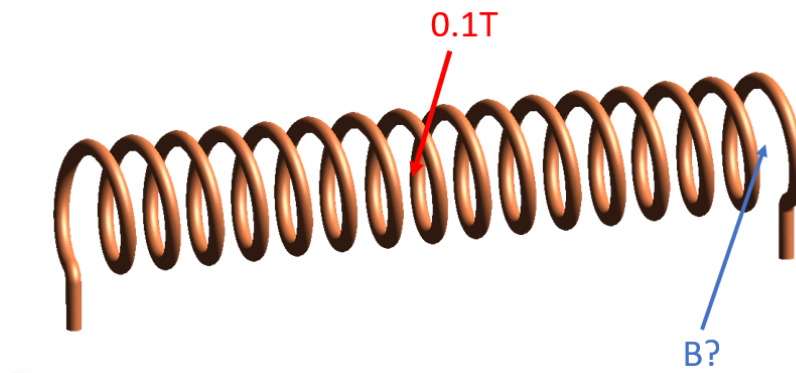
Correct!☐ -4 A☐ 4 A☒ -9 A☐ 9 A☐ None of the above**Question 15****3 / 3 pts**

Consider a toroid carrying a current i that has a square cross-section. The produced magnetic field is largest

☐ at the center of the hole☒ just inside the toroid at its inner surface☐ just inside the toroid at its outer surface☐ at any point inside (the field is uniform)☐ none of the above**Correct!****Question 16****3 / 3 pts**

The magnetic field of a very long solenoid far away from the edges is measured to be 0.1 T. What is the magnetic field right on the edge of the

solenoid (marked with the blue arrow in the drawing below)?



Correct!

- ☐ 0 T
- ☐ 0.1 T
- ☒ 0.05 T
- ☐ -1 T
- ☐ None of the above

Question 17

3 / 3 pts

If you place a drop of liquid near a south pole of a strong bar magnet, the drop of liquid develops a small magnetic field pointing away from the bar. The field goes away when the bar magnet is removed. The liquid is

Correct!

- ☐ Non-magnetic
- ☒ Diamagnetic
- ☐ Paramagnetic
- ☐ Ferromagnetic

☐ No conclusions are possible

Question 18

3 / 3 pts

You measure a magnetic field in a solenoid to be $1 \mu\text{T}$ with a Hall probe. Then, you fill the inner volume of the solenoid with unknown metal and you measure the field to increase by a factor of 1000. When you switch off the current, the metal continues to produce a magnetic field on its own. This metal might very likely contain appreciable amounts of

☐ Copper

☐ Bismuth

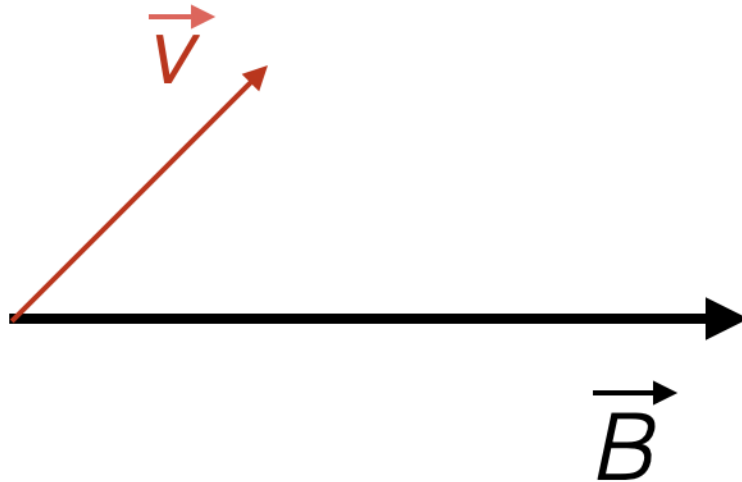
☐ Aluminum

☐ Silver

☒ Iron

Correct!

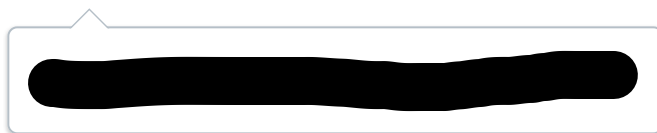
Question 19



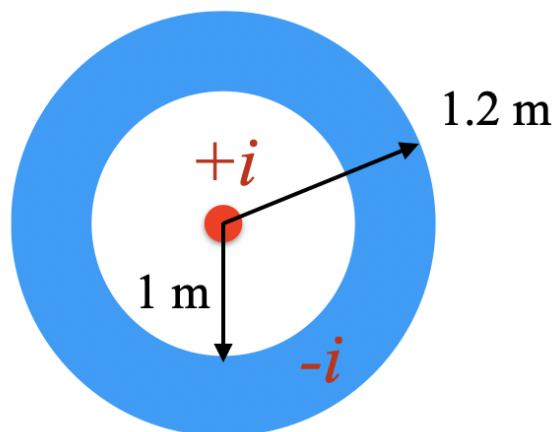
A positively charged particle with a mass $m = 1.7 \times 10^{-27}$ kg and a charge $e = 1.6 \times 10^{-19}$ C moves with a speed $v = 3.0 \times 10^6$ m/s in the direction illustrated by a red arrow in the drawing above. The particle enters a region with a uniform magnetic field of magnitude $B = 0.1$ T (direction is given as a black arrow in the drawing). An angle between velocity and magnetic field vectors is 30 degrees.

- (a) (3 points) Part of the particle's trajectory in this region will be circular motion in a plane transverse to B . Show that the radius of this circle is $R = 0.16$ m.
- (b) (3 points) Show that the time it takes the proton to execute one complete revolution around the circle is $T = 6.7 \times 10^{-7}$ s.
- (c) (3 points) Sketch the three-dimensional trajectory of the particle.
- (d) (3 points) How far will the particle move in the direction of B in the time $T = 6.7 \times 10^{-7}$ s?

[↓ image.jpg \(https://k-state.instructure.com/files/17190115/download\)](https://k-state.instructure.com/files/17190115/download)



Question 20

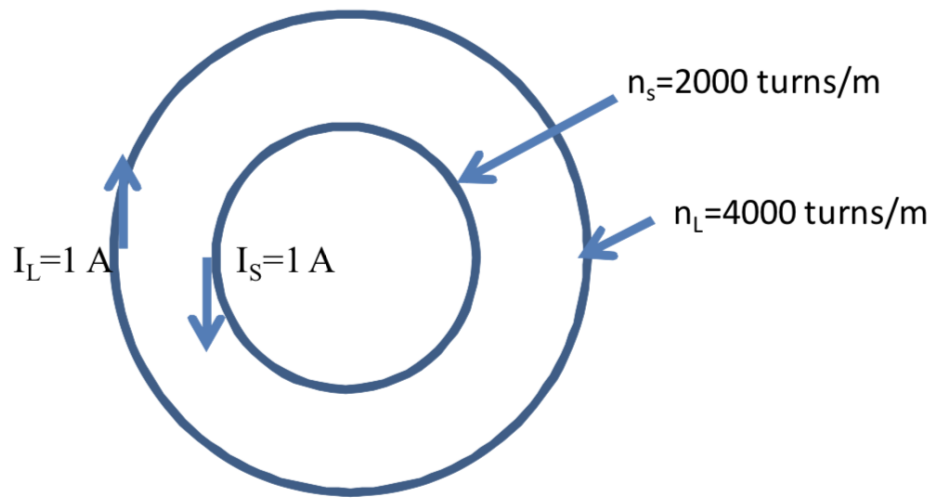


In the image above you see a cross-sectional view of a coaxial cable with a core, depicted as a red circle, that carries a current of 2 A directed out of page/screen (marked as $+i$) and the blue colored shield that has a return current of 2 A in the opposite direction (into the screen/page) marked as $-i$. The shield and the core are conducting and you can assume that the current is distributed uniformly across the surface area of the conductor. The core's size is rather negligible, while the shield's inner radius is 1 m, and the outer radius is 1.2 m.

- (a) (3 points) Determine the magnetic field magnitude at the center of the core ($r = 0\text{ m}$).
- (b) (3 points) Determine the magnetic field magnitude at $r = 0.5\text{ m}$.
- (c) (3 points) Determine the magnetic field magnitude at $r = 1.1\text{ m}$.
- (d) (3 points) Determine the magnetic field magnitude at $r = 2.0\text{ m}$.

↓ [image.jpg \(https://k-state.instructure.com/files/17190126/download\)](https://k-state.instructure.com/files/17190126/download)

Question 21



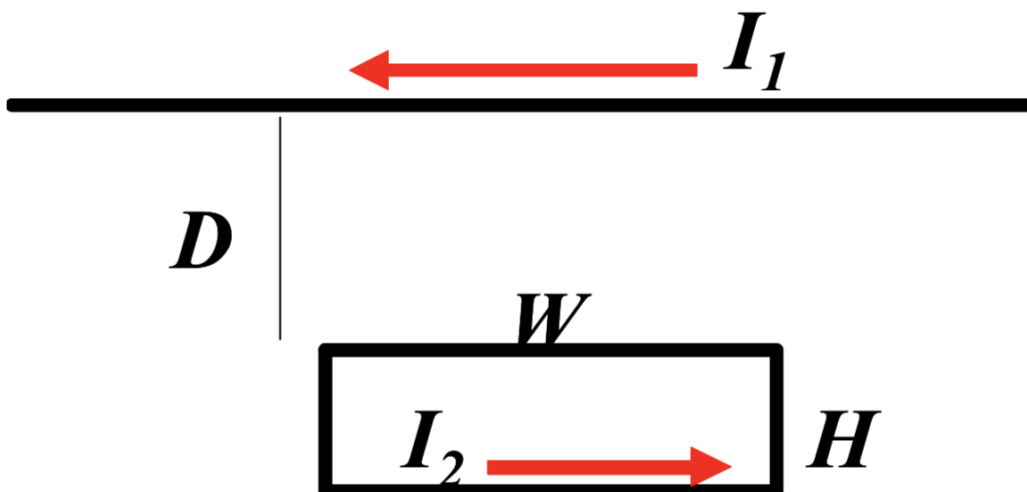
Consider a pair of very long solenoids that share the same axis of symmetry with the smaller diameter solenoid's completely enclosed by the larger diameter's one. In a figure above you can see a cross-sectional view of these two solenoids. The outer one has a wire density of $n_L = 4000$ turns/m, while the inner one has $n_S = 2000$ turns/m. Both solenoids have the same 1 A current, but the winding of the solenoids are opposite to each other, effectively making the current to flow counterclockwise for the inner solenoid and clockwise for the outer.

- (a) (4 points) Find the magnitude of the magnetic field at the center of the two solenoids.
- (b) (4 points) Find the magnitude of the magnetic field in the space between the two solenoid windings.
- (c) (4 points) Determine the magnetic field that would result at the center of the two solenoids if the smaller solenoid were to be filled with a ferromagnet with a relative permeability $\mu = 100$.

↓ [image.jpg \(https://k-state.instructure.com/files/17190135/download\)](https://k-state.instructure.com/files/17190135/download)



Question 22



You have a very long wire with a current I_1 directed to the left and a loop with the width W and a height H and a current I_2 moving in a counter-clockwise direction as illustrated in the figure above. The distance between the wire and the loop is D .

- (a) (4 points) determine the magnitude of the magnetic force on the loop and indicate the direction of that force.
- (b) (2 points) determine the magnetic dipole moment of the loop.
- (c) (4 points) determine the magnitude of the magnetic torque on the loop and indicate which side of the loop will be lifted from the plane of the page/screen (left, right, top, bottom) or show that the torque is zero and neither of the sides will be lifted.



↓ [image.jpg \(https://k-state.instructure.com/files/17190143/download\)](https://k-state.instructure.com/files/17190143/download)

